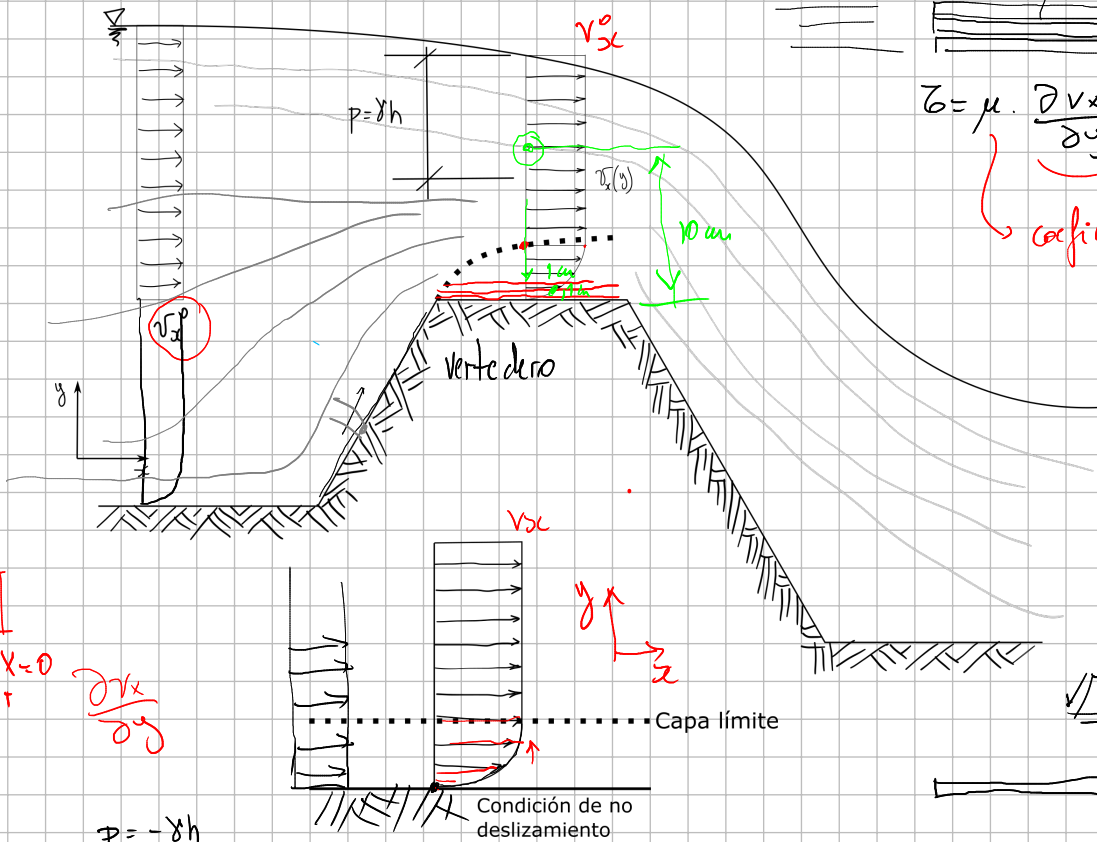
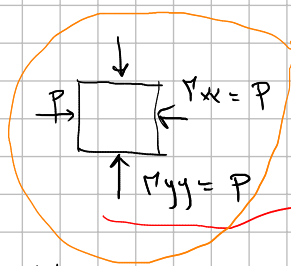


Fluidos viscosos

$\tau = \mu \cdot \frac{\partial v_x}{\partial y}$ Ley de Newton del fluido viscoso
 μ coeficiente de viscosidad



$\nabla \cdot \mathbf{v} + \kappa = \rho \alpha$
 $\nabla \cdot \mathbf{v} = \rho \alpha$
 $\frac{\partial v_x}{\partial y}$



$\tau_{xy} = \tau_{yx}$
 $\tau_{xy} = \mu \cdot \frac{\partial v_x}{\partial y}$ viscosidad

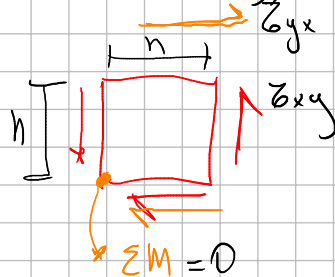
$\tau_{ij} = \tau_{ji}$
 $\tau_{12} = \tau_{21}$

$\mathbf{T} = \begin{bmatrix} -\gamma h & \mu \frac{\partial v_x}{\partial y} & 0 \\ \mu \frac{\partial v_x}{\partial y} & P & 0 \\ 0 & 0 & P \end{bmatrix}$

Fluido newtoniano incompresible

$\tau_{xy} = \mu \cdot \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) = \mu \cdot \frac{\partial v_x}{\partial y}$

$\tau_{yx} = \mu \cdot \left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right) = \mu \cdot \frac{\partial v_x}{\partial y}$



$\sum M = \tau_{xy} \cdot h - \tau_{yx} \cdot h = 0$

$(\tau_{xy} - \tau_{yx}) \cdot h = 0 \quad h \rightarrow 0 \neq 0$
 $\tau_{xy} = \tau_{yx}$

$\sigma_{ij} = -p\delta_{ij} + 2\mu V_{ij}$

Δ (7.3-7)

$\tau_{11} = -p + 2\mu \frac{\partial v_1}{\partial x_1}$
 $\tau_{12} =$

$$\underline{\underline{\sigma}} = \begin{bmatrix} p & \mu \cdot \frac{\partial v_x}{\partial y} \\ \mu \cdot \frac{\partial v_x}{\partial y} & p \end{bmatrix}$$

$$p \neq 0 \text{ y } \alpha \neq 0$$

$$\underline{n} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

Vector tension

$$\underline{T} = \underline{\underline{\sigma}} \cdot \underline{n} = \begin{bmatrix} p & \alpha \\ \alpha & p \end{bmatrix} \cdot \begin{bmatrix} n_x \\ n_y \end{bmatrix} = \begin{bmatrix} p n_x + \alpha n_y \\ \alpha n_x + p n_y \end{bmatrix}$$

hidroestático

$$\|\underline{T}\|^2 = |p|^2$$

$$\begin{aligned} \|\underline{T}\|^2 &= (p n_x + \alpha n_y)^2 + (\alpha n_x + p n_y)^2 \\ &= (p n_x)^2 + 2 p n_x \alpha n_y + (\alpha n_y)^2 + \\ &\quad (\alpha n_x)^2 + 2 \alpha n_x \cdot p n_y + (p n_y)^2 = \\ \|\underline{T}\| &= \sqrt{p^2 + \alpha^2 + 2 \alpha p \cdot \cos(2\theta)} \end{aligned}$$

$$\alpha = 0 \rightarrow \frac{\partial v_x}{\partial y} = 0$$

$$\begin{aligned} &|\underline{T}|(\theta) \rightarrow \\ &|\underline{T}|(\varphi) \end{aligned}$$

